

Complex analysis, Lecture 2

Recall from last time

Let $D \subseteq \mathbb{C}$ be an open disk and $u, v : D \rightarrow \mathbb{R}$ of class C^1 . Then the following are equivalent:

1. $f = u - iv$ is **holomorphic** on D : the complex limit

$$f'(z) = \lim_{h \rightarrow 0} \frac{f(z+h) - f(z)}{h}, \quad h \in \mathbb{C} \setminus \{0\},$$

exists at every $z \in D$;

2. the vector field (u, v) is divergence-free and curl-free:

$$\operatorname{div}(u, v) = 0 \quad \text{and} \quad \operatorname{curl}(u, v) = 0;$$

Recall from last time, continued

3. u and v satisfy the **Cauchy–Riemann equations**

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \quad \text{and} \quad \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = 0.$$

In fluid terms: (u, v) is the velocity field of an **incompressible, irrotational** flow.

A second interpretation: electrostatics

Think of (u, v) as an **electric field** E confined to two dimensions (the cross-section of the field of long charged wires perpendicular to the plane).

Meaning of E . A test charge q placed at (x, y) feels the force $qE(x, y) = q(u(x, y), v(x, y))$: E is the force per unit charge. (For fluids, (u, v) was the velocity of the particle at (x, y) .)

Electrostatics, continued

In electrostatics, in units where $\varepsilon_0 = 1$,

$$\operatorname{div} E = \rho, \quad \operatorname{curl} E = 0,$$

where ρ is the charge density. In a region with **no charge** ($\rho = 0$):

$$\operatorname{div} E = 0 \quad \text{and} \quad \operatorname{curl} E = 0.$$

Exactly the equations of last time: $f = u - iv$ **is holomorphic**.

Here f represents the electric field itself. With the usual electrostatic voltage Φ , the sign convention is $E = -\nabla\Phi$.

Where are the charges?

The condition $\rho = 0$ does **not** force the field to vanish: a constant electric field is an immediate counterexample. A field is determined only after boundary conditions or behaviour at infinity are also prescribed.

The right way to think about it: our holomorphic f is defined on a region D (a disk, say). The charges are **outside** D — on plates and wires surrounding it, at infinity, or at points we have excluded from D . Inside D there is no charge, so $\rho = 0$ there, and that is all the Cauchy–Riemann equations say.

Example: $f(z) = \frac{1}{z - a}$ on a disk D not containing a . On D the field is holomorphic; its source is the point a , outside the region. Example: $f = 1/z$ on $D = \mathbb{C} \setminus \{0\}$ — the charge is at the one point we left out.

Holomorphic functions describe the electric field away from the charges.

What does a charge contribute?

Claim

Let $E_0 := \frac{1}{2\pi} \frac{(x, y)}{x^2 + y^2}$, i.e. $f_0 = u_0 - iv_0 = \frac{1}{2\pi} \cdot \frac{1}{z}$. Then

$$\operatorname{div} E_0 = \delta_0 \quad \text{and} \quad \operatorname{curl} E_0 = 0.$$

Meaning. “ $\operatorname{div} E_0 = \delta_0$ ” is shorthand for: for every region D with smooth boundary, $\oint_{\partial D} E_0 \cdot n \, ds = 1$ if $0 \in D$ and $= 0$ if $0 \notin \bar{D}$ — what the divergence theorem would give for a unit charge concentrated at the origin.

Consequence. Maxwell’s equations are linear, so adding $qf_0 = \frac{q}{2\pi} \cdot \frac{1}{z}$ to f adds $q \delta_0$ to ρ and nothing to $\partial_t B$:

A point charge q at z_0 is the term $\frac{q}{2\pi} \cdot \frac{1}{z - z_0}$ in f . The singularity is the charge.

Verification of the claim

Away from 0. With $r^2 = x^2 + y^2$,

$$2\pi \operatorname{div} E_0 = \frac{\partial}{\partial x} \frac{x}{r^2} + \frac{\partial}{\partial y} \frac{y}{r^2} = \frac{r^2 - 2x^2}{r^4} + \frac{r^2 - 2y^2}{r^4} = \frac{2r^2 - 2(x^2 + y^2)}{r^4} = 0,$$

$$2\pi \operatorname{curl} E_0 = \frac{\partial}{\partial x} \frac{y}{r^2} - \frac{\partial}{\partial y} \frac{x}{r^2} = \frac{-2xy + 2xy}{r^4} = 0.$$

So by the divergence theorem, a region not containing 0 has zero flux.

Verification of the claim, continued

Through a circle $|z| = r$. The outward normal is $(x, y)/r$, so $E_0 \cdot n = \frac{1}{2\pi r}$, and

$$\oint_{|z|=r} E_0 \cdot n \, ds = \frac{1}{2\pi r} \cdot 2\pi r = 1 \quad \text{for every } r.$$

Because E_0 is radial, its circulation around the circle is 0; Green's theorem on the punctured region then shows that its circulation around any enclosing loop is 0, so no curl is concentrated at the origin.

Through any loop around 0. Let $0 \in D$ and remove a small disk $|z| \leq \varepsilon$. On what remains, $\operatorname{div} E_0 = 0$, so the divergence theorem says the flux out of ∂D equals the flux out of the small circle, which is 1. $\square$

Gauss's theorem

Suppose the charge consists of k **point charges** $q_1, \dots, q_k$ at the points $z_1, \dots, z_k$:
 $\rho = \sum_j q_j \delta_{z_j}$, so that $\operatorname{div} E = \sum_j q_j \delta_{z_j}$ and $\operatorname{curl} E = 0$.

Gauss's theorem

Let γ be a positively oriented simple closed curve bounding a region D , not passing through any z_j . Then

$$\oint_{\gamma} E \cdot n \, ds = \sum_{z_j \in D} q_j \quad \text{and} \quad \oint_{\gamma} E \cdot t \, ds = 0.$$

Gauss's theorem: proof

Proof. The divergence theorem and Green's theorem, with $\operatorname{div} E = \rho$ and $\operatorname{curl} E = 0$:

$$\oint_{\gamma} E \cdot n \, ds = \iint_D \operatorname{div} E \, dA = \iint_D \sum_j q_j \delta_{z_j} \, dA = \sum_{z_j \in D} q_j,$$

$$\oint_{\gamma} E \cdot t \, ds = \iint_D \operatorname{curl} E \, dA = 0. \quad \square$$

What Gauss's theorem says about complex functions

For a closed curve γ and $f = u - iv$, define $\oint_{\gamma} f(z) dz := \oint_{\gamma} (u - iv)(dx + i dy)$.

Multiplying out,

$$f dz = \underbrace{(u dx + v dy)}_{E \cdot t ds} + i \underbrace{(u dy - v dx)}_{E \cdot n ds},$$

so by Gauss's theorem,

$$\oint_{\gamma} f dz = \text{circulation} + i \cdot \text{flux} = i \sum_{z_j \in D} q_j.$$

What Gauss's theorem says about complex functions, continued

- ▶ **No charge inside γ** , i.e. f holomorphic on D : $\oint_{\gamma} f(z) dz = 0$. **Cauchy's theorem.**
- ▶ **Charges q_j at $z_j \in D$** , i.e. f holomorphic on D except for terms $\frac{q_j}{2\pi} \cdot \frac{1}{z-z_j}$:
 $\oint_{\gamma} f(z) dz = i \sum_{z_j \in D} q_j$, e.g. $\oint_{\gamma} \frac{dz}{z-z_j} = 2\pi i$. Only the enclosed charges matter, not the shape of γ . **The residue theorem.**

But what about $1/z^2$?

$1/z^2$ is holomorphic on $\mathbb{C} \setminus \{0\}$ and singular at 0, yet

$\oint_{|z|=r} \frac{dz}{z^2} = \int_0^{2\pi} \frac{ire^{i\theta}}{r^2 e^{2i\theta}} d\theta = \frac{i}{r} \int_0^{2\pi} e^{-i\theta} d\theta = 0$: by Gauss, the **net charge** at 0 is zero.

What kind of singularity has zero total charge but a nonzero field?

A dipole. Put a charge $+q$ at δ and $-q$ at 0, and let $\delta \rightarrow 0$ with $q\delta = 1$ fixed:

$$\frac{q}{2\pi} \left(\frac{1}{z-\delta} - \frac{1}{z} \right) = \frac{q}{2\pi} \cdot \frac{\delta}{z(z-\delta)} \longrightarrow \frac{1}{2\pi} \cdot \frac{1}{z^2}.$$

Two opposite charges, infinitely close and infinitely strong: total charge 0, but a nonzero dipole charge distribution (a derivative of a delta) and hence a field nonetheless. Likewise $1/z^3$ is a quadrupole, and so on.

The moral: multipoles and the residue

The moral. A singularity of f at z_0 can be a whole combination of multipoles,

$$\frac{a_1}{z - z_0} + \frac{a_2}{(z - z_0)^2} + \frac{a_3}{(z - z_0)^3} + \cdots,$$

and the contour integral sees only the first coefficient:

$$\oint_{\gamma} f \, dz = 2\pi i a_1.$$

The **residue** is a_1 . If $a_1 = \alpha + i\beta$, then the net charge (flux) is $2\pi\alpha$ and the circulation is $-2\pi\beta$. For purely electric point charges a_1 is real, so the enclosed charge is $2\pi a_1$.

u and v are harmonic

In both interpretations, u and v are very special functions. Assume $u, v \in C^2$ and take the Cauchy–Riemann equations

$$u_x + v_y = 0, \quad v_x - u_y = 0.$$

Apply ∂_x to the first and ∂_y to the second, then **subtract**:

$$(u_{xx} + v_{yx}) - (v_{xy} - u_{yy}) = u_{xx} + u_{yy} = 0.$$

Apply ∂_y to the first and ∂_x to the second, then **add**:

$$(u_{xy} + v_{yy}) + (v_{xx} - u_{yx}) = v_{xx} + v_{yy} = 0.$$

u and v are harmonic, continued

Definition

A C^2 function h is **harmonic** if $\Delta h := h_{xx} + h_{yy} = 0$.

So u and v are harmonic. Note that $\Delta h = 0$ is the **heat equation** $\partial_t h = \Delta h$ in two dimensions, for a temperature distribution h that no longer changes in time: harmonic functions are equilibrium temperatures.

Harmonic functions are determined by their boundary values

Theorem

Let Ω be a bounded open set with piecewise C^1 boundary, and let $h_1, h_2 \in C^2(\overline{\Omega})$ be harmonic on Ω . If $h_1 = h_2$ on the boundary $\partial\Omega$, then $h_1 = h_2$ on all of Ω .

Physically obvious for temperatures: hold the boundary of a plate at a prescribed temperature and wait; the equilibrium temperature inside is whatever it is, and nothing else. We prove it by the **energy method**.

Proof: the energy method

Let $w = h_1 - h_2$: harmonic on Ω , C^2 up to the boundary, and zero on $\partial\Omega$.

Step 1: a product rule. For the vector field $w \nabla w = (w w_x, w w_y)$,

$$\begin{aligned}\operatorname{div}(w \nabla w) &= \partial_x(w w_x) + \partial_y(w w_y) \\ &= (w_x^2 + w w_{xx}) + (w_y^2 + w w_{yy}) = |\nabla w|^2 + w \Delta w.\end{aligned}$$

Step 2: the divergence theorem applied to $w \nabla w$ on Ω :

$$\iint_{\Omega} \operatorname{div}(w \nabla w) dA = \oint_{\partial\Omega} w \nabla w \cdot n ds = 0,$$

the boundary integral vanishing because $w = 0$ at every point of $\partial\Omega$.

The energy method, continued

Step 3: combine. Integrating Step 1 over Ω and using Step 2,

$$0 = \iint_{\Omega} |\nabla w|^2 dA + \iint_{\Omega} w \Delta w dA = \iint_{\Omega} |\nabla w|^2 dA, \quad \text{since } \Delta w = 0.$$

Step 4: conclude. The integrand $|\nabla w|^2$ is continuous and ≥ 0 with integral 0, so $\nabla w \equiv 0$ on Ω . Hence w is constant on each connected component of Ω ; each constant is 0 because $w = 0$ on that component's boundary. $\square$

The quantity $\iint_{\Omega} |\nabla w|^2 dA$ is the **energy** of w — the kinetic energy of the fluid, or the energy stored in the electric field. The proof says: a harmonic function with zero boundary values has zero energy, hence is zero.

In practice: how does one compute the solution?

We know the boundary values determine the harmonic function. But given the boundary values, how do we *find* it?

$$\Delta u = 0 \text{ in } \Omega, \quad u = g \text{ on } \partial\Omega.$$

The standard tool is the **finite element method**. Three steps:

1. rewrite the problem in **weak form**;
2. choose a finite-dimensional space of functions built on a grid;
3. solve the resulting linear system.

Next lecture. At the end of it we will have a concrete, finite-dimensional picture of *why* the boundary values determine everything.